What Investment Data Implies about the AI Transition

Quant Finance Academy

Wharton, October 28, 2026

The question

How much will AI contribute to economic growth?

- Existing answers span orders of magnitude
- Acemoglu (2024): TFP $\uparrow$ $\sim$0.7 pp over ten years
- Amodei (2024, 2026): "multiple trillions a year by 2028 or 2029" of infrastructure spending; decades of scientific progress in years

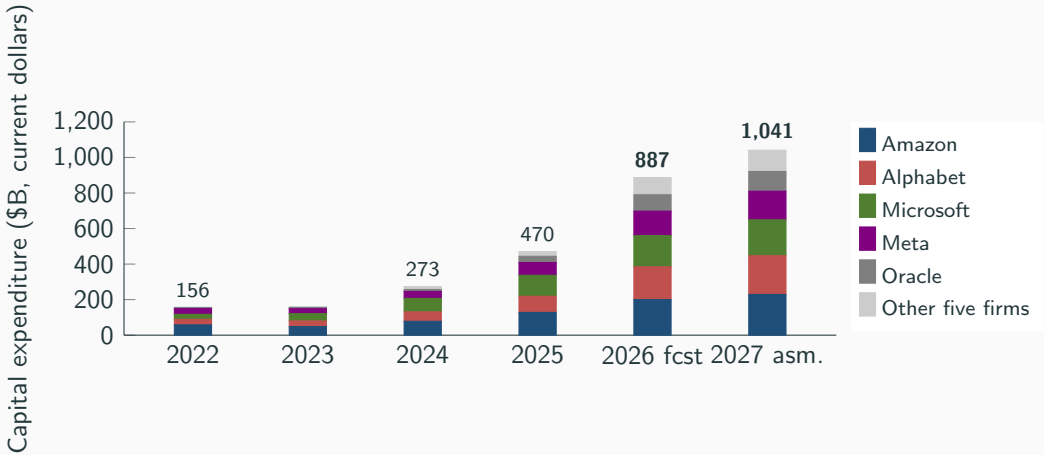
We use quantitative techniques to answer this question.

The approach

Facts	Ten firms' filings: capex \$156B (2022) $\rightarrow$ \$782B (2026 forecast), in 2022 dollars
Model	Firms invest until the marginal dollar has zero NPV
Calibration	Capital built through 2027 reveals the boom: AI productivity $\times 3.3$
Scenarios	Two coin flips at $p = 0.5$ in 2028–29; the value of capital in place
Growth	GDP paths to 2032 and to 2050

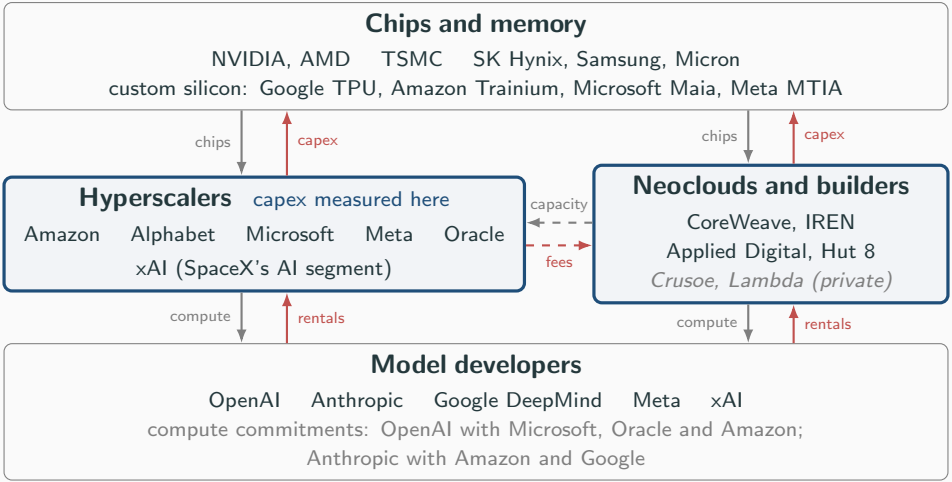
What the firms are spending

Ten firms' capital expenditure



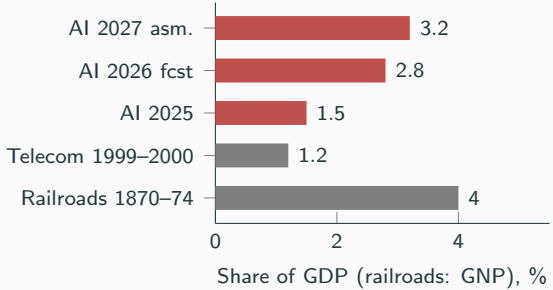
Flat in 2022–23; up by two-thirds in 2024 and again in 2025; the 2026 forecast nearly doubles 2025. In 2022 dollars: 156, 153, 255, 427, 782, 917. Other five firms: xAI, CoreWeave, IREN, Applied Digital, Hut 8. 2027 holds each firm's net addition to capital at its 2026 level; sources in the backup.

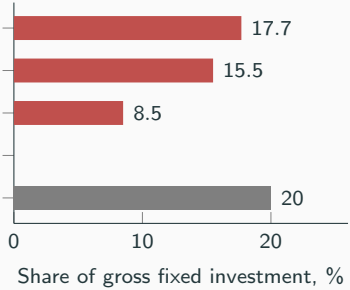
Where the ten firms sit in the AI supply chain



Capacity sold by a neocloud to a hyperscaler is a service contract, an operating expense for the buyer, so the equipment is counted once. Crusoe and Lambda publish no figures and are excluded (paper, Appendix A).

Two historical benchmarks





Railroads: gross railroad capital expenditure (Ulmer 1960) over nominal GNP (Gallman and Rhode 2020), cumulative 1870–74, about 4%; peak years of the early 1870s reach 5–6%. About one-fifth of gross domestic fixed capital formation.
Telecom: BEA private fixed investment in communication equipment and structures over GDP.
AI: ten-firm capex (worldwide, current dollars) over U.S. GDP and gross private fixed investment; 2026–27 GDP and investment extended at 3% nominal growth.

A model of investment under rare booms

Two sectors, rare productivity boom

Two sectors $j \in \{a, n\}$ (AI, non-AI), Cobb–Douglas:

$$Y_{jt} = z_{jt}^{1-\alpha} K_{jt}^{\alpha}, \qquad K_{j,t+1} = (1 - \delta_j) K_{jt} + I_{jt}.$$

z_{jt} productivity, K_{jt} capital, I_{jt} investment, α the capital share, δ_j depreciation.

AI productivity subject to a rare boom (probability p_t):

$$z_{a,t+1} = \begin{cases} z_{at}\, e^{\xi} & \text{w.p. } p_t \\ z_{at} & \text{w.p. } 1 - p_t \end{cases} \qquad \xi > 0.$$

Non-AI unaffected.

Timing: I_{at}, and so $K_{a,t+1}$, is chosen at t knowing z_{at} and p_t but not $z_{a,t+1}$.

Parameters: $\alpha = 1/3$, $\delta_a = 0.25$ (hardware replaced in about four years), $\delta_n = 0.07$. Output is gross of wages, $z_j = A_j^{1/(1-\alpha)} \bar{L}_j$ with $\bar{L}_j$ fixed; investment is the same net of wages (paper, Appendix B).

The value of the firm: a present discounted value

AI firms maximize the present value of output net of investment, discounted at the return r_a investors require:

$$V_{at} = \max_{\{I_{as}\}_{s \geq t}} \mathbb{E}_t \sum_{s=t}^{\infty} \frac{1}{(1+r_a)^{s-t}} \left[z_{as}^{1-\alpha} K_{as}^{\alpha} - I_{as} \right], \qquad K_{a,s+1} = (1-\delta_a) K_{as} + I_{as}.$$

Open economy: global investors supply capital at $r_a = 0.08$ (a 2% real risk-free rate plus a 6% equity premium), so r_a is given.

One more dollar of capital at t costs \$1 of investment today. Next year it produces its marginal product, $\alpha \left(z_{a,t+1} / K_{a,t+1} \right)^{1-\alpha}$, and $1 - \delta_a$ dollars of it are still in place. Its contribution to value:

$$-1 \; + \; \frac{1}{1+r_a} \, \mathbb{E}_t \left[\alpha \left(\frac{z_{a,t+1}}{K_{a,t+1}} \right)^{1-\alpha} + (1-\delta_a) \right].$$

Invest until this is zero: **the marginal dollar has zero NPV**.

Optimal investment

First-order condition for $K_{a,t+1}$: expected marginal product equals the user cost,

$$\mathbb{E}_t\left[\alpha\left(\frac{z_{a,t+1}}{K_{a,t+1}}\right)^{1-\alpha}\right] = r_a + \delta_a.$$

At $r_a = 0.08$ and $\delta_a = 0.25$: each dollar of AI capital must earn 33 cents a year.

Evaluate the expectation over the boom; the bracket is the **boom option value** $\Omega(p_t)$:

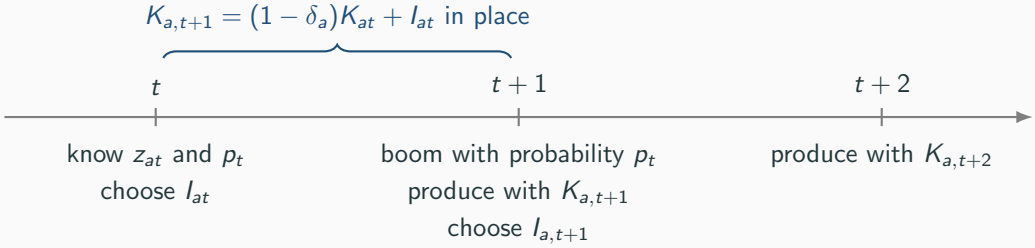
$$\alpha\left(\frac{z_{at}}{K_{a,t+1}}\right)^{1-\alpha} \underbrace{\left[(1-p_t) + p_t\, e^{(1-\alpha)\xi}\right]}_{\equiv\,\Omega(p_t)\,\geq\,1} = r_a + \delta_a.$$

Optimal capital chosen at time t:

$$\boxed{K^*_{a,t+1} \;=\; z_{at}\left(\frac{\alpha\,\Omega(p_t)}{r_a + \delta_a}\right)^{1/(1-\alpha)}.}$$

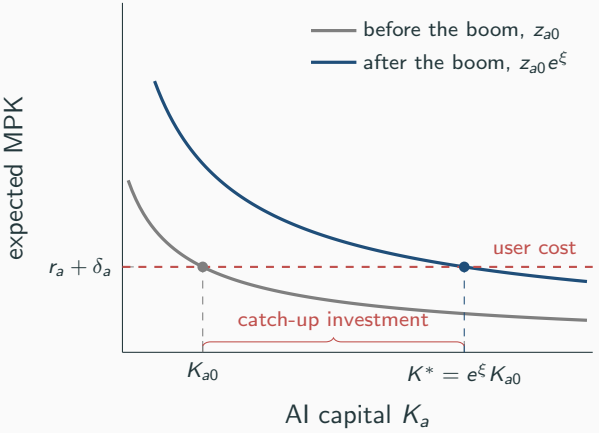
Non-AI: the same with $\Omega = 1$ (no boom exposure) and $r_n = 0.061$, which matches 2022 GDP.

Timing: capital is chosen before the boom is known



- Capital in place at $t+1$ earns $e^{(1-\alpha)\xi}$ times as much if the boom arrives. Its expected marginal product carries the factor $\Omega(p_t) = (1-p_t) + p_t\, e^{(1-\alpha)\xi}$
- **Option value:** a rise in p raises investment before any boom occurs
- In the calibration below: in 2028, p rises from ≈ 0 to 0.5 with no boom, and target capital rises from \$2,048B to \$4,162B

Invest until the marginal dollar is zero NPV



- Capital is added until expected MPK equals the user cost, $r_a + \delta_a$
- A boom shifts the schedule; with p unchanged, the ratio of capital stocks is e^{ξ}
- With economic profits, the same investment implies a larger boom

Revealed preference: ξ from investment data

Proposition (capital ratio after a boom)
Suppose a boom occurs at t and p shifts $p_{t-1} \to p_t$. Then

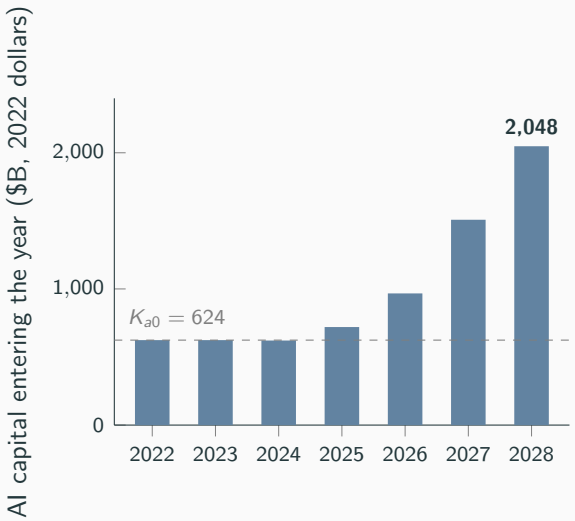
$$\frac{K^*_{a,t+1}}{K^*_{at}} = e^{\xi} \left(\frac{\Omega(p_t)}{\Omega(p_{t-1})} \right)^{1/(1-\alpha)}.$$

Key: the ratio is independent of r_a.

- $r_a + \delta_a$ cancels in the ratio
- $\Rightarrow$ identify ξ from investment data
- $\Rightarrow$ no stand on the required return

What the investment implies

From investment to the capital stock



Initial steady state:

$$K_{a0} = I_{a,2022}/\delta_a = 156/0.25 = \$624\text{B}$$

Capex data, 2023–2027:

$$K_{a,t+1} = (1 - \delta_a)K_{at} + I_{at}$$

Capital at its target entering 2028:

$$e^{\hat{\xi}} = \frac{K_{a,2028}}{K_{a0}} = \frac{2{,}048}{624} \approx 3.28$$

$\hat{\xi} \approx 1.19$. Assumes zero economic profits and no change in discount rates. The five largest firms alone: \$1,859B, a factor of 2.98.

What the 3.3 describes

- Productivity of the AI sector, z_a: a permanent jump
- With capital adjusted to the boom (entering 2028), the sector's capital, output, and earnings are each 3.3 times their 2022 level
- The sector is 2.4% of output before the boom
- It is what managers believe, as revealed by investment at zero NPV

Is it large?

- Against the technology's own progress since 2022, a factor of 3.3 may be modest
- For the economy: the AI sector at 7.5% of output and GDP 5.6% higher, with no further boom and no spillovers

The firms' company-wide earnings are a separate question; the paper takes no stance.

What if managers are wrong?

The revealed-preference $\hat{\xi}$ is what managers *believe*. Wedges to the true gain:

True gain smaller $\hat{\xi}$ too high

- **Mirage / bubble.** Late-1990s fiber precedent; customers will not pay for more AI usage $\Rightarrow$ NPV-negative capex, defaults, disinvestment
- **Misattributed p.** Capex that anticipates a rise in p is read as boom size: $\hat{\xi}$ too high, eventual capital unchanged

True gain larger $\hat{\xi}$ too low

- **Under-investment.** Managerial risk aversion, anchoring, the quiet life
- **Option to buy distressed rivals' capital** below replacement cost reduces the incentive to build now
- Plus the measurement items: firms outside the ten, non-AI capital in the 2022 baseline, purchases on credit, zero profits

Winner-take-all races: $\hat{\xi}$ unaffected (per firm, $\hat{\xi}_i = 0.77$–1.91); only the eventual level of capital is lower.

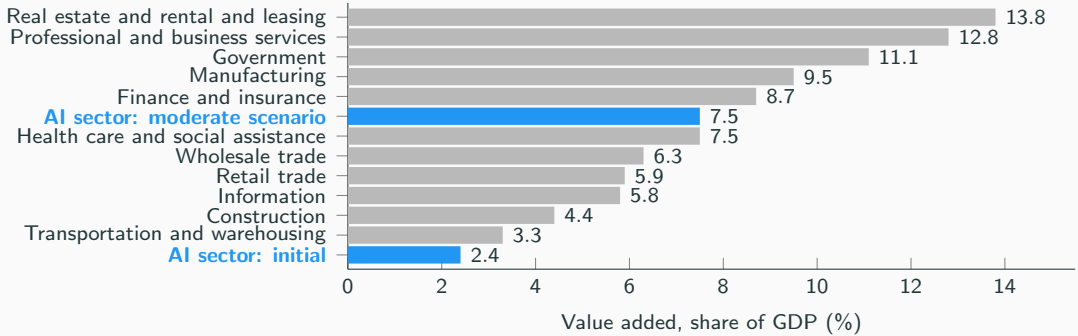
The first boom alone: before and after

	Initial	After, no further boom
AI capital ($B)	624	2,179
AI investment ($B per year)	156	545
AI-sector output ($B)	618	2,070
AI share of output (%)	2.4	7.5
GDP		+5.6%

2022 dollars. Capital fully adjusted at the long-run boom probability (3.5% a year, set below), hence $2,179B rather than $2,048B; non-AI output held fixed (no spillovers).

Task-based estimates: TFP up ∼0.7 pp over ten years (Acemoglu 2024).

The AI sector at 7.5% of output



Industries: BEA value added by industry as a percentage of GDP, 2025. AI sector: model share of output, with non-AI output held at its initial level; the ten firms' activity is spread across BEA industries. Transformative scenario: 21.1%; singularity: 46.7%.

Value added is an industry's own wages and capital income: for health care, what providers keep after paying their suppliers, 7.5% of GDP, against national health expenditure of 18% (2024). The model's AI share is value added.

Scenarios for 2028–2030

Scenario design: a window of elevated p

Structural break in the boom probability:

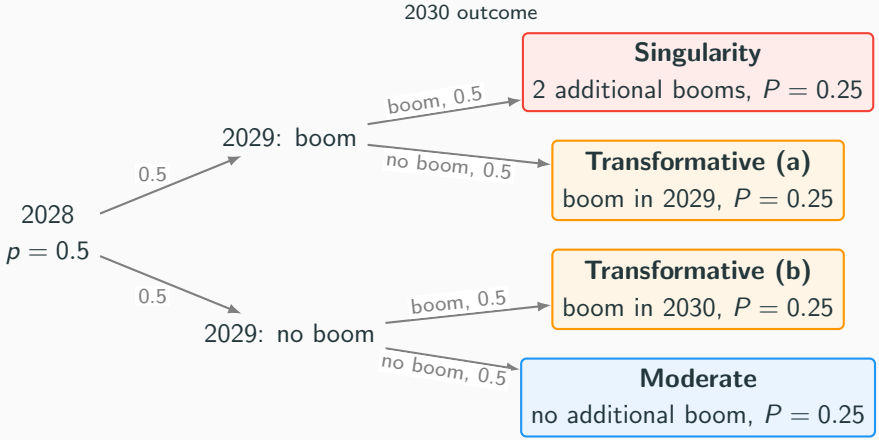
$$p_t = \begin{cases} p_0 \approx 0 & t < 2028 \\ p_{\max} = 0.50 & t \in \{2028, 2029\} \\ p_* & t > 2029 \end{cases}$$

Two i.i.d. draws at $p_{\max} = 0.50$, governing booms in 2029 and 2030:

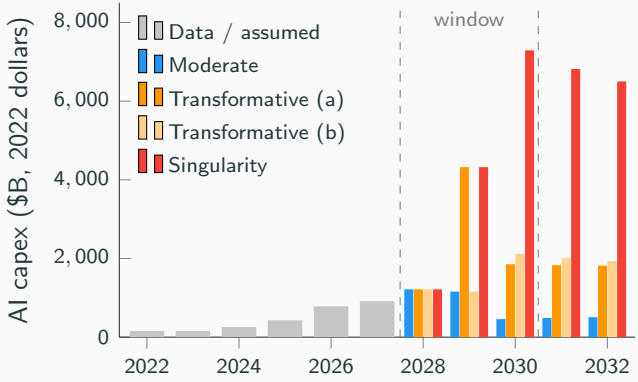
Scenario	Add'l booms	Probability
Moderate	0	0.25
Transformative	1	0.50
Singularity	2	0.25

$p_{\max} = 0.5$ gives maximal uncertainty (a Bernoulli variable has maximal variance at 0.5). The rise in p is unanticipated through 2027.

Four scenarios from two coin flips



Investment paths by scenario



2028: p rises to $p_{\max}$, no boom yet. Option value lifts the target stock from \$2,048B to \$4,162B. Same investment in every scenario.

Partial adjustment from 2028: $K_{t+1} = K_t + (K^* - K_t)/\theta_a$, $\theta_a = 3$

Post-window ($\delta_a K^*$): \$545B / \$1,788B / \$5,869B per year (mod./trans./sing.)

U.S. gross private fixed investment, 2022: \$4.7T.

Booms after 2030: how often can they come?

After the window, booms of the same size arrive with probability p_* a year. Expected AI-sector productivity growth:

$$g = p_* \, (e^{\xi} - 1).$$

A firm is worth the present value of its cash flows. If they grow at g and investors discount at r,

$$\text{Value} = \frac{\text{Cash flow}}{r - g}, \qquad \text{finite only if } g < r.$$

For the AI sector this puts a ceiling on p_*:

$$p_* < \frac{r_a}{e^{\xi} - 1} = \frac{0.08}{2.28} = 0.0351 \quad \Longrightarrow \quad \boxed{p_* = 0.035}, \qquad g \approx 8.0\% \text{ a year}$$

AI raises long-run growth by an amount the investment data do not determine; we take the largest value consistent with finite firm values. At $p_* = 0.034$ the paths to 2050 are nearly the same (expected GDP gain 30% vs. 32% in the moderate scenario, 348% vs. 363% in the singularity).

The scenarios against industry statements

Industry statement	Model, dated	Mod.	Trans.	Sing.
Amodei (Feb. 2026): infrastructure spending of "multiple trillions a year by 2028 or 2029"	Investment in 2030, $T	0.5	2	7.3
SpaceX S-1 (May 2026): goal of 100 GW of compute launched a year, "as early as 2028"; sustained, a stock of 400 GW for SpaceX alone	Post-window stock of AI capital, GW	62	203	666
Musk (May 2026): "the economy is probably twice its current size in 5, maybe 6 years"	GDP in 2032, multiple of initial	1.06	1.23	1.73
Amodei (2026): a "country of geniuses in a datacenter" around 2028; one reading, 300M effective workers on top of the sector's 13M: $\times 24$	Effective AI-sector labor after the window, multiple of 2027	1	3.3	10.8

Partial-adjustment paths ($\theta_a = 3$), transformative path (a), 2022 dollars; GW are post-window steady-state stocks at $40B per GW (2026 dollars). GDP is the AI contribution alone. Each boom multiplies effective AI-sector labor by $e^{\xi} \approx 3.3$.

GDP growth

Post-window snapshot

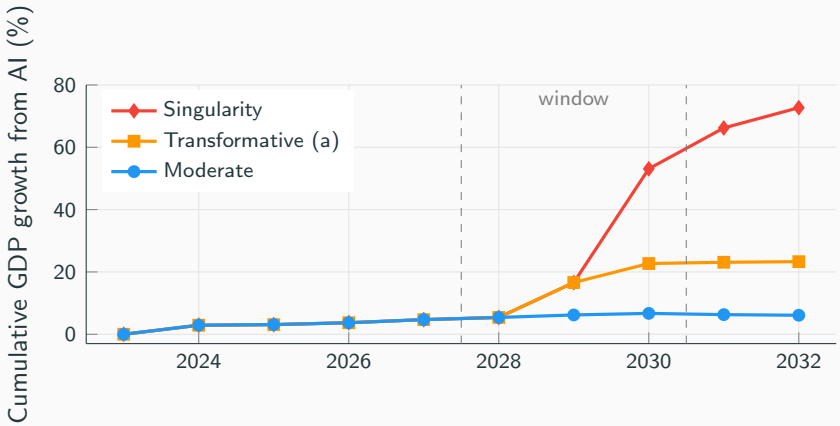
	AI share s_a of output (%)	GDP gain (%)	Post-window I_a ($B/yr)
Initial	2.4	—	156
Moderate	7.5	5.6	545
Transformative	21.1	23.7	1,788
Singularity	46.7	83.2	5,869

Steady state at scenario-terminal z_a with $K_a = K_a^*(z_a, p_*)$, no booms after 2030; non-AI output fixed (no spillovers); 2022 dollars. A one-time increase in the level of GDP, excluding normal-times growth.

Long-run expected AI-sector growth (aggregate growth from AI: s_a times this):

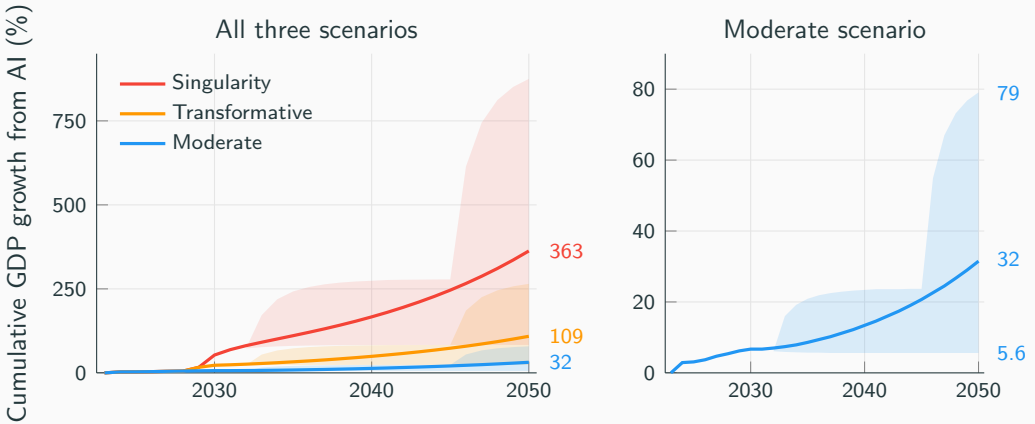
$$g = p_*(e^{\xi} - 1) = 0.035 \times 2.28 \approx 8.0\%/\text{yr}, \qquad \text{in rare discrete jumps.}$$

GDP transition path (2024–2032)



Capex data through 2027; partial adjustment ($\theta_a = 3$) from 2028. In 2030: 6.7%, 23%, 53%; capital fully adjusted: 5.6%, 24%, 83%. The first boom is placed at an instant (as Barro 2006 does for disasters); little of it is in measured GDP through mid-2026.

Long-run distributions to 2050



Booms continue at $p_* = 0.035$ a year from 2031. Lines: expected paths. Bands: 10th–90th percentiles, exact over the 2^{20} boom histories. Partial adjustment throughout; transformative is path (a).

Two observations about the long run

1. **With no spillovers, the AI share is first order.** Aggregate growth from AI is the sector's growth weighted by its share; 7.5% vs. 47% after the window is what compounds into 32% vs. 363% expected by 2050.

2. **The number of booms after 2030 matters as much as the number in the window.** The scenarios' 10th–90th percentile bands overlap: the moderate scenario's 90th percentile, 79%, nearly reaches the singularity scenario's 10th, 83%.

The moderate scenario is already large: one boom and nothing further in the window gives 5.6 pp once capital has adjusted, and 32 pp in expectation by 2050.

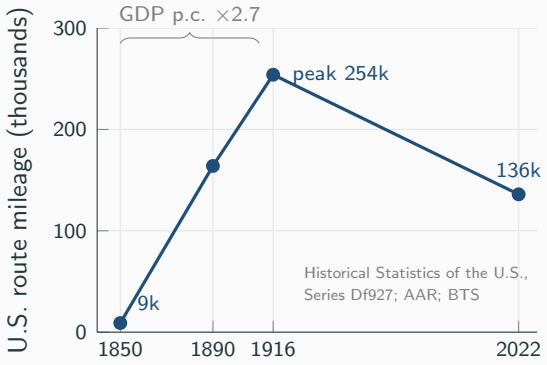
Historical context and conclusion

Growth episodes on one scale

Episode	Period	Years	Multiplier: GDP per capita	Multiplier: AI sector
British Industrial Revolution	1760–1840	80	1.4	—
U.S. railroad era	1850–1910	60	2.7	—
Second Industrial Revolution (U.S.)	1870–1920	50	2.1	—
Japan postwar; Korea; Taiwan; Singapore; China	1945–2020	25–30	4–9	—
U.S. IT boom	1995–2005	10	1.3	—
AI moderate (1 boom)	2024–2030	6	1.1	3.3
AI transformative (2 booms)	2024–2030	6	1.2	10.8
AI singularity (3 booms)	2024–2030	6	1.5	35.4
AI moderate, booms continue at p_*	2030–2060	30	1.6	10.0
AI transformative, booms continue at p_*	2030–2060	30	2.7	10.0
AI singularity, booms continue at p_*	2030–2060	30	5.8	10.0

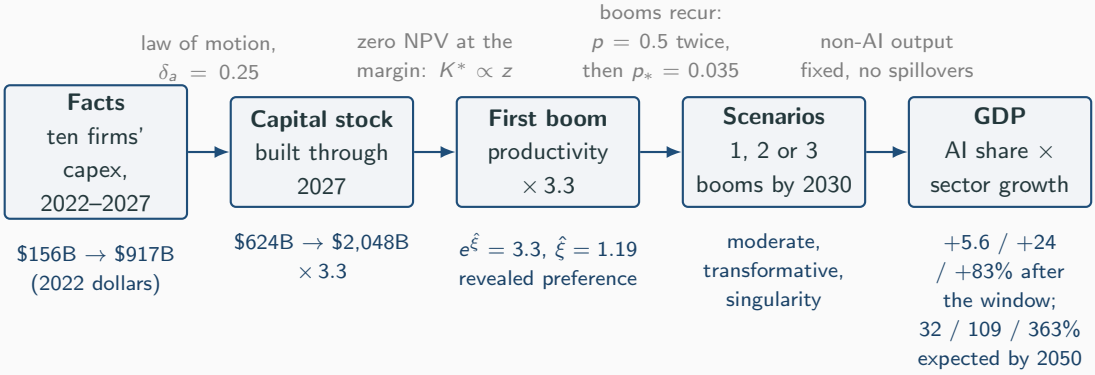
Historical rows: real GDP per capita, Maddison Project Database 2023 (Japan 4.2, Korea 9.0, Taiwan 7.3, Singapore 5.7, China 5.8); they include growth from every source. AI rows: the contribution of AI alone, non-AI output fixed; window rows are the 2030 values on the transition paths, 30-year rows expected GDP in 2060 relative to 2030.

The railroad analogue: overshooting came decades after the gain



1. Route miles 9,000 $\rightarrow$ 254,000; GDP per capita $\times 2.7$ over the railroad era
2. 46% of peak mileage abandoned, but decades later, with the rise of competing transportation
3. The productivity gain was real; the decline was a later shock to the sector's z
4. The question is whether the boom justifies the investment when made: at $\delta_a = 0.25$, today's hardware is largely replaced within four years

The argument in one picture



The GDP paths are conditional on the revealed-preference estimate, which assumes that the firms driving the buildout are, in aggregate, not destroying value.

Thank you

Questions?

Backup: capital expenditure by firm

	2022	2023	2024	2025	2026 fcst	2027 asm.	2027 proj.
Amazon	58	48	78	128	200	229	388
Alphabet	31	32	53	91	185	218	400
Microsoft	28	41	76	118	175	202	300
Meta	32	28	39	72	138	161	248
Oracle	7	7	11	35	92	111	348
xAI (SpaceX AI segment)	—	—	6	13	55	68	
CoreWeave	—	3	9	10	27	33	
IREN	—	—	—	2	7	9	
Applied Digital	—	—	1	1	6	7	
Hut 8	—	—	—	—	2	3	
Total, current dollars	**156**	**159**	**273**	**470**	**887**	**1,041**	
Total, 2022 dollars	**156**	**153**	**255**	**427**	**782**	**917**	

$ billions; calendar years (Oracle, Applied Digital: December–November); each firm's own measure of capex. 2026: management forecasts for the four largest, year to date plus extrapolation for the others. 2027 assumed: 2026 capex plus depreciation on the 2026 addition, in 2026 dollars; projected: the latest quarter's growth continued. Paper, Table 1.

Backup: relative to the macroeconomy

	2022	2023	2024	2025	2026 fcst	2027 asm.
AI capex ($B, current)	156	159	273	470	**887**	**1,041**
U.S. nominal GDP ($T)	26.1	27.8	29.4	30.9	∼31.8	∼32.7
U.S. gross priv. fixed inv. ($T)	4.7	5.0	5.3	5.6	∼5.7	∼5.9
Share of GDP (%)	0.6	0.6	0.9	1.5	**2.8**	**3.2**
Share of priv. fixed inv. (%)	3.3	3.2	5.2	8.5	**15.5**	**17.7**

GDP and gross private fixed investment: BEA (NIPA Table 1.1.5), extended at 3% nominal growth for 2026–27. Capex is consolidated worldwide; GDP and investment are U.S. only, so to the extent spending occurs abroad the ratios overstate the U.S. shares. Paper, Table 1, Panel B.

Backup: the same calculation firm by firm

Firm	2022 capex ($B)	$K_{i,2028}$ ($B)	$e^{\hat{\xi}_i}$	$\hat{\xi}_i$
Amazon	58	500	2.16	0.77
Meta	32	324	2.53	0.93
Alphabet	31	421	3.39	1.22
Microsoft	28	424	3.78	1.33
Oracle	7	190	6.78	1.91
Five largest firms	156	1,859	2.98	1.09
All ten firms	156	2,048	3.28	1.19

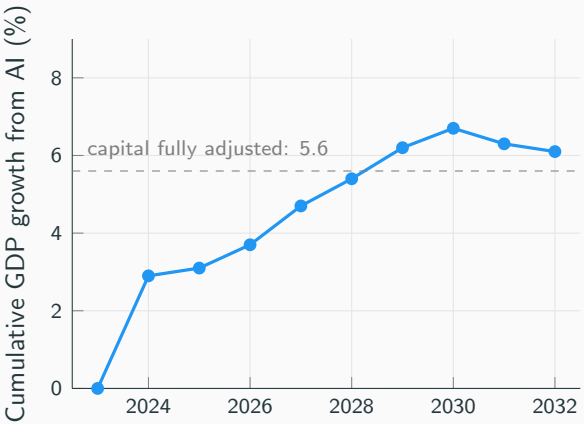
Each firm as if it were the sole survivor: high on a per-firm basis too. 2022 dollars; the other five firms had no capital in 2022, so the aggregate multiplier is an average of the five firms' weighted by their 2022 capital, plus the other five firms' capital. Paper, Section 3.4 and Appendix A.4.

Backup: how sensitive is 3.3?

	$e^{\hat{\xi}}$	
Baseline: capital entering 2028 over initial capital, $\delta_a = 0.25$, ten firms	**3.28**	
Lower		
$\delta_a = 0.15$ (mean life 6.7 years rather than 4), re-estimated	2.44	⇓
Cumulative 2024–27 investment, adjustment placed in 2024 (June WP)	2.61	⇓
2027 decision already at boom probability $p_{2027} = 0.10, 0.25, 0.50$	2.84, 2.47, 2.14	⇓
Five largest firms only	2.98	⇓
Higher		
Firms outside the ten (Crusoe, Lambda); purchases on credit until paid		⇑
Part of 2022 capex is non-AI capital		⇑
Economic profits; investment restrained to preserve market power		⇑

Depreciation: paper, Appendix B. Anticipation and cumulative-investment rows: the paper's code, not reported in the paper. r_a cancels in the ratio.

Backup: the first boom alone, the path of GDP



- Moderate scenario (no boom after the first): about 7% in 2030, then back toward 5.6%. Capital built in 2028–29, when the boom probability is high, exceeds the long-run optimum
- The model places the boom at an instant (as Barro 2006 does for disasters)
- Little of it is in measured GDP through mid-2026. The Solow history (1987) is the precedent for a lag between the investment and the statistics

Backup: how a boom translates to output

A boom at $t+1$: productivity jumps, capital is predetermined $\Rightarrow$ two-stage response.

Impact period $(t+1)$**:** only the productivity factor moves

$$Y_{a,t+1} \;=\; e^{(1-\alpha)\xi}\, z_{at}^{1-\alpha}\, (K^*_{a,t+1})^{\alpha}.$$

At $\alpha = 1/3$, $\xi \approx 1.19$: $e^{(1-\alpha)\xi} = 2.21$ $(+121\%)$.

Adjustment period $(t+2)$**:** capital catches up, $K^*_{a,t+2} = e^{\xi} K^*_{a,t+1}$

$$Y_{a,t+2} \;=\; e^{\xi}\, z_{at}^{1-\alpha}\, (K^*_{a,t+1})^{\alpha}.$$

Full factor $e^{\xi} = 3.28$ realized; capital contributes the remaining $e^{\alpha\xi} = 1.49$.

α governs the split between immediate and deferred output response.

Backup: from capital paths to GDP growth

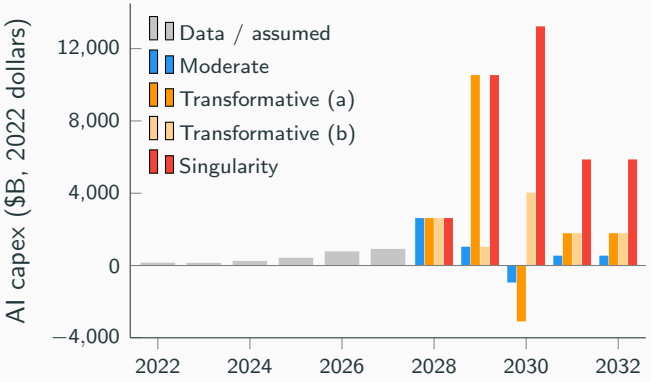
AI output along each scenario path; non-AI output held at its initial level Y_{n0} (no spillovers):

$$Y_{at} = z_{at}^{1-\alpha} K_{at}^{\alpha}, \qquad z_{at} = z_0\, e^{n_t \xi} \quad (n_t = \text{booms realized by } t, \text{ incl. the first boom}).$$

Cumulative GDP growth from AI: $\dfrac{Y_{at} + Y_{n0}}{\text{GDP}_0} - 1$, $\text{GDP}_0 = Y_{a0} + Y_{n0}$. AI capital:

- through 2027: $K_{a,t+1} = (1 - \delta_a) K_{at} + I_{at}$, with I_{at} from the capex data
- from 2028: $K_{t+1} = K_t + (1/\theta_a)(K^* - K_t)$, $\theta_a = 3$
- from 2031 (long-run distributions): booms i.i.d. at $p_* = 0.035$; partial adjustment continues
- AI share of output: $s_a = Y_a/(Y_a + Y_n)$

Backup: frictionless investment paths



- 2028: K^* rises from \$2,048B to \$4,162B; I_{2028} = \$2,625B in every scenario
- 2029: \$10,537B with a boom, \$1,040B without
- 2030: negative where p reverts to p_* with no new boom

Expected GDP gain by 2050:
partial adjustment 32 / 109 / 363%
full adjustment 35 / 119 / 396%

Backup: a lower depreciation rate, a higher required return

	AI share (%)	GDP gain (%)	I_a ($B/yr)
Baseline: $\delta_a = 0.25$, $r_a = 0.08$, $e^{\hat{\xi}} = 3.28$, $p_* = 0.035$			
Initial	2.4	—	156
Moderate	7.5	5.6	545
Transformative	21.1	23.7	1,788
Singularity	46.7	83.2	5,869
$\delta_a = 0.15$, *re-estimated* $e^{\hat{\xi}} = 2.44$, $p_* = 0.035$			
Moderate	6.6	4.1	397
Transformative	14.6	13.9	970
Singularity	29.5	37.9	2,368
$r_a = 0.15$ *(industry hurdle rate)*, $\delta_a = 0.25$, $e^{\hat{\xi}} = 3.28$, $p_* = 0.035$			
Moderate	9.0	6.8	545
Transformative	24.5	28.7	1,788
Singularity	51.6	100.8	5,869

Longer-lived capital is cheaper to hold, so a given stock is evidence of less productivity. A higher r_a leaves 3.28 unchanged but raises output per dollar of capital, $(r_a + \delta_a)/\alpha$. Paper, Appendix B.

Backup: a "country of geniuses in a datacenter" through the model

Amodei (2024, 2026): millions of instances smarter than a Nobel Prize winner at 10–100× human speed, around 2028.

Treat it as effective labor: $z_a = A_a^{1/(1-\alpha)} \bar{L}_a$ rises in proportion to $\bar{L}_a$.

- One reading: 10 million instances at 30× speed = 300 million effective workers
- AI-sector effective workforce: 2.4% of the 167M U.S. labor force = 4M before the boom; ×3.3 = 13M by end-2027
- $300/13 \approx 24$, against $e^{2\hat{\xi}} \approx 10.8$ for the singularity scenario's two further booms

Well beyond the singularity scenario: industry treats further booms as the median outcome. (Paper, Section 3.4 and Appendix B.)